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A theory of inference compute scaling: reasoning through directed stochastic skill search

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Large language models (LLMs) require considerable computation and energy resources during training and deployment. While scaling laws for training have guided much recent progress, inference costs represent a significant and growing component of the overall resource burden, particularly for reasoning models. Existing compute-optimality characterizations that consider model size, dataset size and inference tokens in isolation or fixed combinations may overlook more efficient operating points. We introduce directed stochastic skill search (DS3), a general framework that represents inference as stochastic traversal over a learnt skill graph. From a simplified yet expressive instantiation, we derive closed-form expressions for task success and compute cost across a wide range of inference strategies—including chain-of-thought (CoT) and tree-of-thought (ToT)—enabling comparisons by task difficulty and model capability. We extend a prior graph framework of LLM training to include inference and bridge DS3 with empirical scaling laws. We theoretically recover observed patterns, including linear accuracy scaling with log-compute, variation in preferred inference strategies by task and capability, emergent behaviour elicited by reasoning despite parameter plateaus and both best-of- N and majority voting (MV) captured within one framework. By characterizing training-inference

interdependencies, our framework deepens theoretical understanding and supports principled algorithmic design and resource allocation.

This article is part of the discussion meeting issue 'Bits, neurons and qubits for sustainable AI'.

1. Introduction

Large language model (LLM) task performance improves by scaling the computation spent on learning (i.e. training compute), driven by scaling up both model size and dataset volume [1,2]. This scaling produces progressively richer and more structured internal representations, enabling models to better capture abstract concepts and generalize to unseen tasks [3]. As a result, the training compute used for frontier models has grown by an estimated 4–5× per year [4]. Concurrently, algorithmic improvements have shifted the compute–performance frontier [5], reducing the physical compute required to reach a given performance level by 2–6× annually [6,7]. Performance has also been shown to scale with computation spent on reasoning (i.e. inference compute, also referred to as test-time compute, search or thinking). Therefore, reasoning models have been adopted by industry leaders. As inference-time compute is scaled, a model's capacity to dynamically traverse and leverage its learnt internal structure may be enhanced.

Yet, the performance gains from inference scaling come at a cost. While training involves a substantial but largely one-time expenditure, inference uses energy throughout the model's operational life. For frontier LLMs, inference compute often substantially outweighs training compute [8,9]. The total burden of inference is likely to continue to grow regardless of reducing the cost per token and task by hardware, architectural and algorithmic improvements. Such efficiency gains may ultimately drive greater inference demand, fuelled both by enhanced capabilities and by broader sociotechnical dynamics [10], following the Jevons paradox that greater efficiency leads to increased overall consumption [11].

With usage increases, the cumulative cost of inference may grow substantially [12]. This is further amplified by the inference scaling paradigm, where costs grow nonlinearly as task lengths increase. Nonetheless, compared to training, inference has historically attracted less attention with respect to energetic costs [13]. Compute scaling during both training and inference shapes model capabilities and forms a deeply interdependent system. Energy costs, architectural decisions and performance trade-offs all arise from these coupled dynamics. A unified perspective is essential to guide resource allocation, research priorities and policy frameworks towards sustainable artificial intelligence (AI) grounded in the co-evolution of training and inference.

Here, we introduce a *directed stochastic skill search* (DS3) framework for LLM inference to explain empirical scaling phenomena. DS3 provides closed-form expressions for accuracy and cost, enabling principled selection and design of inference strategies based on task difficulty and model capability. DS3 characterizes trade-offs between sequential and parallel reasoning strategies and provides theoretical grounding for sampling-based approaches such as majority voting (MV), offering interpretations of observed empirical trends. The framework integrates with existing techniques for analysing LLM training, supporting joint analysis of training and inference. This yields a unified characterization of the training-inference system for both design and optimization. The construction recovers known scaling patterns under sequential inference, including linear accuracy gains with logarithmic increases in training and inference compute. We emphasize opportunities for efficiency gains through adaptive inference strategies and training-inference co-design. A unified formulation of training and inference can inform both technical and policy guidance to support sustainable development across the LLM lifecycle.

2. Preliminaries: scaling and compute optimality

As a foundation for our framework, we review selected results and limitations in scaling law theory and compute optimality.

Many scaling laws predict pretraining loss, not task accuracy. While early work showed power-law loss decay with scale of both model size and training dataset size [1,2,14], these formulations required empirical fits. Recent theoretical work has sought to explain their origins [15,16]. Emergent and plateauing behaviours in LLMs—where performance improves sharply or stagnates with small changes in scale—have been observed [17,18]. While their origin remains debated [19], recent theoretical work suggests such transitions result from structured learning dynamics, including the acquisition of discrete knowledge units, or quanta, proportional to model size [16,20]. Skill-based graph representations and task composition studies [21,22] further support this view, aligning with cognitive science findings on learning transitions [23]. However, pretraining loss does not directly predict downstream accuracy. This has led to two-step forecasting methods that fit loss, then map it to accuracy using a function such as a sigmoid [24]. Accuracy can plateau or decline even as loss improves with some tasks [25,26].

Inference scaling aims to improve downstream task performance, which is only indirectly captured by pretraining loss. In fact, increasing inference compute can harm task accuracy even as cross-entropy loss improves, highlighting a mismatch between loss and actual performance. Optimizing directly for pass@ k —a metric that reflects inference-scaled accuracy in sampling-based settings—yields better alignment between training objectives and task performance [27]. As inference compute grows, loss alone offers limited guidance, motivating a focus on task-level accuracy. Chain-of-thought (CoT) reasoning, where models generate intermediate steps before answering, can be scaled either through explicit training [28–30] or simple prompting (e.g. ‘think step by step’) [31]. CoT induces emergent capabilities on tasks with flat or inverse scaling, shifting trends towards positive or stable performance [32,33]. In fact, it has been demonstrated that treating reasoning as a random walk over clustered CoT states—a latent reasoning graph—can improve model performance via data augmentation, compared to direct supervised fine-tuning. This lends support to the view that graph-based structure plays a meaningful role in the reasoning process itself [34]. This underscores the interdependence between training and inference scaling paradigms and the need for their joint optimization.

Training has been the primary focus for improving LLM performance owing to both technical and economic factors. However, LLMs may yield increasingly diminishing gains with scaling training owing to the limited availability of high-quality, unique data [35], as well as the growing cost of achieving order-of-magnitude improvements [36]. In contrast, scaling inference offers a more sample-efficient way to improve [29,30]. A useful distinction in this context is between sequential and parallel inference scaling. Sequential inference scaling involves deepening the model’s reasoning within a single inference trace (e.g. CoT or iterative refinement), often involving longer or more structured reasoning traces. In contrast, parallel inference scaling involves generating multiple independent samples (e.g. best-of- N (BoN), which necessitates selecting among outputs or MV). Hybrid approaches, such as tree-of-thought (ToT), combine elements of both—replicating and refining multiple reasoning paths. Their trade-offs and implementation barriers are distinct, and the most effective balance between them depends on the base model capabilities, task difficulty and computational constraints.

Applying inference scaling to LLMs has faced several challenges. Early base models lacked sufficient generative quality, making it more effective to invest compute in improving outputs rather than generating multiple weaker ones. In many early applications, generating a correct answer was no harder than verifying one. Sequential reasoning methods faced limitations, including restricted context length, attention instability over long sequences and bias accumulation [37–40]. In addition, the open-ended nature of token generation made parallel search largely intractable [28]. Simple techniques like MV or iterative refinement [41–43] have been available, but their modest accuracy gains often do not justify the additional compute for all tasks [44]. Verifier

models have shown promise under outcome and process supervision [45], but often suffer from overfitting and high computational cost [28,46,47]. Training is a one-time cost, while inference scales with usage—making it more efficient at scale to invest in training to reduce per-inference cost. Real-time user expectations further discourage deeper inference strategies.

Recent advances have reshaped the viability of inference scaling. Improvements in base model quality, long-context handling and reasoning stability have made outputs more coherent and reliable, boosting the effectiveness of inference-based strategies. At the same time, tasks like scientific reasoning, coding and long-horizon planning have grown in complexity, widening the gap between generation and verification. These tasks benefit from multi-step reasoning, where users will tolerate slower inference for higher-quality results. More structured generation enables focused search over outputs, and reinforcement learning enables unified models that handle both generation and evaluation [28]—so no need for separate verifiers. Together, these shifts position inference scaling as a practical and increasingly central tool for improving LLM performance. They also call for a re-evaluation of what it means for a system to be compute optimal.

Compute-optimal scaling laws describe how to allocate model components—such as parameters, training dataset size and inference tokens—under a fixed compute budget to maximize performance. These laws resemble allometric scaling in biology, where evolutionary pressures yield systematic scaling relationships—such as brain-to-body mass ratios across species [48–50]. In biology, evolutionary pressures produced divergent scaling laws for brain-to-body mass relationships among hominids, non-human primates and other mammals [50]. Analogously, scaling laws for LLMs are not static: they change with technical progress and altering incentives [51–53]. In particular, the rise of inference scaling exerts novel pressures, causing deviations from previously established compute-optimal trajectories.

Before the rise of inference scaling, performance optimization focused primarily on training compute. Foundational work proposed minimizing pretraining loss under fixed compute budgets, leading to the Chinchilla framework, which recommends training smaller models on more data for better performance [1,2]. Empirical and theoretical results support a fixed ratio between model size and training tokens [2,18]. These approaches typically ignore inference cost. Studies show that, under a fixed inference-compute budget, smaller models with more inference tokens can outperform larger models with fewer, prompting joint optimization of model size and inference strategy [54,55]. However, prior work has relied on models that are training-compute-optimal or close to it—an assumption that is suboptimal. Smaller, overtrained models can yield better performance at lower inference cost [24,56], even beyond 10 000 tokens per parameter [51], though generalization is not always guaranteed [57].

Recent work has emphasized the importance of characterizing compute optimality not only for training but also for the full lifecycle of LLMs [51]. This preference for inference-efficient models leads to scaling strategies that deviate from the Chinchilla-optimal ratio, improving both total compute efficiency and ease of deployment [24]. Minimizing the pretraining loss under a total-compute constraint, without attention costs, has previously been solved numerically by assuming an expected lifetime token output, reducing costs compared to Chinchilla scaling [51]. Yet, this does not consider task performance and omits explicit inference scaling. In practice, algorithmic choices and hardware characteristics further shape the cost structure, altering the resulting optimal allocations [51,53–55]. The framework introduced in this work naturally supports joint optimization of model size, training dataset size and inference strategy to improve performance over a distribution of tasks, under flexibly defined cost constraints.

3. Directed stochastic skill search (DS3)

We refer to the following general inference formalism, along with its concrete instantiations under various representational assumptions, as DS3. At its core, DS3 treats inference as a dynamic ensemble of sequential traces, where each trace i represents a single walk through a latent skill graph governed by stochastic transitions and outcomes. Formally, each trace i unfolds on a directed

graph $\mathcal{G}^{(i)} = (\mathcal{V}^{(i)}, \mathcal{E}^{(i)})$, which may differ across traces. That is, the graph structure itself—its set of vertices and edges—is indexed by i , allowing heterogeneous inference topologies even within a shared ensemble. The vertex set $\mathcal{V}^{(i)} = \mathcal{S}^{(i)} \cup \{\text{BRANCH}^{(i)}\} \cup \{\text{IDLE}^{(i)}\} \cup \{\text{STOP}^{(i)}\}$ consists of a (finite or countably infinite) set of latent skills $\mathcal{S}^{(i)}$ learnt during training, along with distinguished control nodes: $\text{STOP}^{(i)}$, an absorbing vertex that terminates the trace; $\text{IDLE}^{(i)}$, which allows the trace to persist without traversal; and $\text{BRANCH}^{(i)}$, which initiates the creation of one or more new traces. The edge set $\mathcal{E}^{(i)} \subseteq \mathcal{V}^{(i)} \times \mathcal{V}^{(i)}$ defines the allowable transitions between vertices, including any self-loops and transitions into the distinguished nodes.

The process begins from an initialization $\Theta_0^{(i)}$, which encodes prior information available before inference begins—such as the initial prompting, interaction history or previously generated context. At each discrete step $t \in \{1, 2, \dots\}$, a trace is located at node $X_t^{(i)} \in \mathcal{V}^{(i)}$ and emits an outcome $O_t^{(i)}$ from an outcome space $\mathcal{O}^{(i)}$, which may correspond to generated tokens, latent updates or decisions. The corresponding sequences of visited nodes and emitted outcomes are denoted by $\mathbf{X}_t^{(i)} = (X_1^{(i)}, \dots, X_t^{(i)})$ and $\mathbf{O}_t^{(i)} = (O_1^{(i)}, \dots, O_t^{(i)})$, respectively. We define the history of trace i at time t as $\mathbf{H}_t^{(i)} = (\Theta_0^{(i)}, \mathbf{X}_t^{(i)}, \mathbf{O}_t^{(i)})$, which captures all information observed along that trace up to time t . The tuple of all such history across the ensemble of all N_t traces instantiated up to time t is denoted by $\mathcal{H}_t = (\mathbf{H}_t^{(i)})_{i=1}^{N_t}$. At any time t when a trace reaches $\text{BRANCH}^{(i)}$, it may fork into new traces that inherit the parent's history, $\mathbf{H}_t^{(i)}$, at the moment of branching. In particular, for each child trace i' , the history is initialized as $\mathbf{H}_t^{(i')} = \mathbf{H}_t^{(i)}$, and the graph structure is cloned: $\mathcal{G}^{(i')} = \mathcal{G}^{(i)}$. Any trace that does not arise through branching is explicitly defined in the framework, with an associated initialization $\Theta_0^{(i)}$ and graph $\mathcal{G}^{(i)}$, even if it begins in $\text{IDLE}^{(i)}$. The tuple of all graphs used in an inference strategy is $\mathfrak{G} = (\mathcal{G}^{(i)})_{i=1}^{N_t}$.

We define a dynamic external environment state $\mathfrak{E}_t$, which may influence the inference process in real time, adapting to factors such as user interventions or resource availability (e.g. compute, energy and memory). Inference proceeds with each trace evolving independently or according to a collective policy. The walk is governed by two stochastic components. First, a transition kernel $\iota_t^{(i)}$ defines the probability of moving to the next node, conditioned on the history tuple and the environment, and second, a conditional distribution $p_t^{(i)}$ governs outcomes at each step:

$$\Pr[X_t^{(i)} = v \mid \mathcal{H}_{t-1}, \mathfrak{E}_{t-1}] = \iota_t^{(i)}(v; \mathcal{H}_{t-1}, \mathfrak{E}_{t-1}), \quad O_t^{(i)} \sim p_t^{(i)}(\cdot \mid \mathcal{H}_t, \mathfrak{E}_t), \quad v \in \mathcal{V}^{(i)}. \quad (3.1)$$

Next, we explicitly define the dynamic evaluation and cost sets:

$$\mathcal{Q}_t = \left\{ q_t^{(j')}(\mathcal{O}_t, \mathfrak{E}_t) \right\}_{j' \in \mathcal{J}'}, \quad \mathcal{C}_t = \left\{ c_t^{(j)}(\mathcal{H}_t, \mathfrak{E}_t) \right\}_{j \in \mathcal{J}}. \quad (3.2)$$

Each index $j' \in \mathcal{J}'$ denotes an evaluation function $q_t^{(j')}$, applied to the tuple of outcomes $\mathcal{O}_t = (\mathbf{O}_t^{(i)})_{i=1}^{N_t}$ across all traces. These functions provide semantic or task-level assessments of inference quality—such as correctness, coherence or goal satisfaction—and may also vary with the environment $\mathfrak{E}_t$. This set can include verifiers, pass@k metrics or other evaluative procedures. Each index $j \in \mathcal{J}$ corresponds to a cost function $c_t^{(j)}$, which may depend on individual trace histories or aggregate ensemble behaviour via $\mathcal{H}_t$, as well as on the external environment $\mathfrak{E}_t$. These costs capture both local metrics, such as per-step compute or latency, and global ones like inter-trace contention or coordination, modulated by real-time constraints such as energy pricing or hardware efficiency.

The termination time $\mathcal{T} \in \mathbb{N}$ is the first time step at which all traces have reached their respective stop nodes: $\mathcal{T} = \min \left\{ t \in \mathbb{N} \mid X_t^{(i)} = \text{STOP}^{(i)} \text{ for all } i \in \{1, \dots, N_t\} \right\}$. Inference proceeds over steps $t \in \{1, \dots, \mathcal{T}\}$, after which no further transitions or outcomes occur.

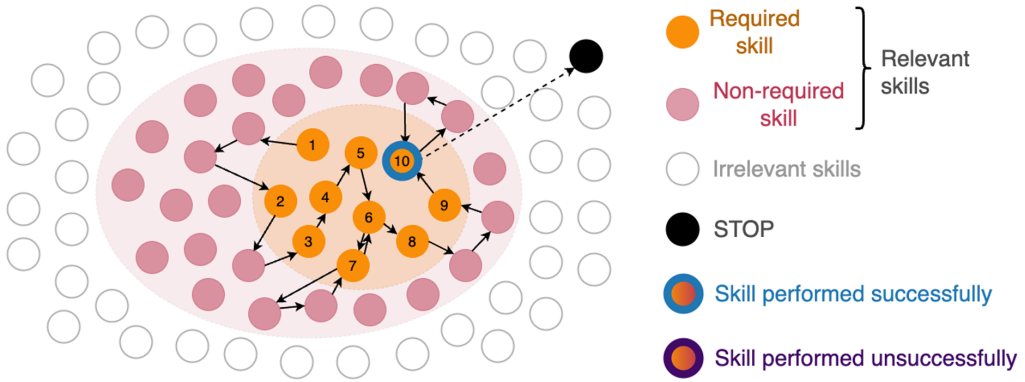


Figure 1. An example DS3 trace for a CoT policy, terminating on the final required skill (after 22 steps).

(a) Idealized set-up

We analyse four idealized inference strategies: CoT, ToT(1), BoN and MV. ToT(1) is a simplified variant of ToT: after each skill step, the model evaluates all active traces, selects one to continue and branches it into b new traces. This process repeats at every step. These idealized settings allow us to isolate the core dynamics of inference under a mathematically simple framework that can then be extended to a vast family of inference strategies. Figure 1 illustrates a representative non-branching trajectory through the skill graph. To facilitate tractable analysis, we impose several structural simplifications on the general DS3 framework. All traces evolve over the same fully connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and act on a shared skill set $\mathcal{S}$, partitioned into three disjoint subsets: required skills $\mathcal{S}_{\mathcal{R}^+}$, relevant-but-not-required skills $\mathcal{S}_{\mathcal{R}^-}$ and irrelevant skills $\mathcal{S}_{\mathcal{I}}$. Task completion requires sequential execution of m skills (possibly with repetitions) drawn from $\mathcal{S}_{\mathcal{R}^+}$. The complete relevant set is $\mathcal{S}_{\mathcal{R}} = \mathcal{S}_{\mathcal{R}^+} \cup \mathcal{S}_{\mathcal{R}^-}$, with cardinality $M = |\mathcal{S}_{\mathcal{R}}|$.

We introduce a directionality coefficient $\delta \in [0, 1]$, where $\delta = 0$ corresponds to a fully undirected random walk and $\delta = 1$ corresponds to a perfectly directed walk toward the next required skill. Let s^* denote the next required skill in the task sequence. We define $\mathbb{1}_{\text{ST}}(v_{t-1}) = 1$ if one of the following conditions holds at step $t - 1$: (i) all m required skills have been completed in the correct order; (ii) the inference budget is exhausted; or (iii) the current trace was not selected following a BRANCH and the outcome of the subsequent step. Otherwise, $\mathbb{1}_{\text{ST}}(v_{t-1}) = 0$. We also define $\mathbb{1}_{\text{BR}}(v_{t-1}) = 1$ if the trace at step $t - 1$ initiates branching (i.e. under ToT(1)), and 0 otherwise. Transitions occur between vertices $v_{t-1}, v_t \in \mathcal{S}_{\mathcal{R}} \cup \{\text{BRANCH}\} \cup \{\text{STOP}\}$, and we define the corresponding transition kernel as

$$\iota(v_t | v_{t-1}) = \begin{cases} 1, & \text{if } v_t = \text{STOP}, \mathbb{1}_{\text{ST}}(v_{t-1}) = 1, \text{ or if } v_t = \text{BRANCH}, \mathbb{1}_{\text{BR}}(v_{t-1}) = 1, \\ \delta + (1 - \delta) \frac{1}{M}, & \text{if } v_t = s^*, \mathbb{1}_{\text{BR}}(v_{t-1}) = 0, \mathbb{1}_{\text{ST}}(v_{t-1}) = 0, \\ (1 - \delta) \frac{1}{M}, & \text{if } v_t \in \mathcal{S}_{\mathcal{R}} \setminus \{s^*\}, \mathbb{1}_{\text{BR}}(v_{t-1}) = 0, \mathbb{1}_{\text{ST}}(v_{t-1}) = 0. \end{cases} \quad (3.3)$$

Then, $\hat{\iota} = \delta + (1 - \delta) \frac{1}{M} \in \left[\frac{1}{M}, 1 \right]$, hereafter referred to as the directionality, quantifies the model's ability to traverse its latent skill space effectively. A higher value of $\hat{\iota}$ indicates a stronger bias towards the correct next step in the skill sequence.

We define a simplified outcome space $\mathcal{O} = \{(0, \omega), (1, \omega), \emptyset\}$, where each skill application emits a constant ω tokens and returns a binary outcome: success $(1, \omega)$ or failure $(0, \omega)$ in applying the next required skill s^* . The symbol $\emptyset$ denotes a null outcome emitted at control nodes such as

BRANCH or STOP. The corresponding observation kernel is defined as

$$p(O_t | v_t) = \begin{cases} p, & \text{if } O_t = (1, \omega), v_t = s^*, \\ 1 - p, & \text{if } O_t = (0, \omega), v_t = s^*, \\ 1, & \text{if } O_t = (0, \omega), v_t \in \mathcal{S}_{\mathcal{R}}, \text{ or if } O_t = \emptyset, v_t \in \{\text{STOP}, \text{BRANCH}\}. \end{cases} \quad (3.4)$$

Here, $p \in [0, 1]$ is the fixed probability of successfully applying any visited skill. We assume p is uniform across all skills in $\mathcal{S}_{\mathcal{R}}$ and independent of the trace history. Assuming perfect memory benefits inference strategies with longer sequential reasoning chains by ensuring intermediate failures or detours do not erase prior progress. In contrast, when incorporating penalties for longer chains (e.g. through context length constraints), our assumption of strictly sequential completion can disproportionately hinder sequential strategies: a required skill contributes to task completion only if it is executed at the correct position in the sequence. However, we shall revisit the effects of memory and context length when comparing sequential and parallel strategies. For clarity, we also do not impose any maximum context length in this idealized framework.

We consider compute as the sole cost, defined as the cumulative sum of per-trace compute costs. We omit batching effects, memory usage and hardware variability for simplicity. We further assume that verification is handled by oracle verifiers, which always select the best available trace. This removes the need to represent verification effort explicitly: evaluation reduces to scanning all traces and selecting the one with the highest cumulative outcome score. Let $q_t(\mathbf{O}_t)$ denote the number of successful outcomes by time t , defined as: $q_t(\mathbf{O}_t) = \sum_{\nu=1}^t \mathbb{1}\{O_\nu = (1, \omega)\}$. We denote the prompt token budget by Ω_{prompt} and take it as the initialization $\Theta_0^{(i)} = \Omega_{\text{prompt}}$ for all N_1 active traces at $t = 1$. The context length at time t for trace i is $n_{\text{ctx}}^{(i)}(\Omega_0, \mathbf{O}_{t-1}^{(i)}) = \Omega_0 + \sum_{\nu=1}^{t-1} \omega \cdot \mathbb{1}\{O_\nu^{(i)} \in \{(0, \omega), (1, \omega)\}\}$, which accumulates over all emitted tokens in the trace history. As final components of the set-up, we define the evaluation set and the total inference cost:

$$\mathcal{Q}_t = \{q_t^{(i)} := q_t(\mathbf{O}_t^{(i)})\}_{i=1}^{N_t}, \quad C_{\text{inf}} = \sum_{t=1}^{\mathcal{T}} \sum_{i=1}^{N_t} \sum_{\omega=1}^{\omega} C_{\text{forward}}\left(n_{\text{ctx}}^{(i)}(\Omega_{\text{prompt}}, \mathbf{O}_{t-1}^{(i)})\right). \quad (3.5)$$

Here, each $q_t^{(i)}$ is the cumulative number of required skills correctly applied by trace i up to time t , and C_{inf} is the total inference cost, aggregated over all time steps, active traces and token-level forward passes. The per-step context length $n_{\text{ctx}}^{(i)}$ is passed as input to the forward pass cost function C_{forward} , which yields the compute cost of a single inference step.

Table 1 lists a summary of the accuracy and compute costs of the inference strategies examined in this work. This analysis provides a foundation for comparing the structural trade-offs of inference strategies in relation to compute constraints and success probabilities. While we focus on a representative subset of policies, the framework readily extends to a broader space of strategies, enabling principled analysis of their computational costs and effectiveness.

(b) Balance between sequential and parallel reasoning

We examine a case study that illustrates how the structure of inference affects performance, contrasting CoT with ToT(1), under a fixed compute constraint. Assuming a constant per-step success probability $\hat{p}$ independent of step index t , a fully sequential method is preferred when the total number of generated tokens is held fixed. However, under a fixed compute budget, the cost structure differs: because attention scales quadratically with context length, parallel strategies can often generate a larger number of tokens for the same total compute. In the large inference budget regime, the quadratic attention terms dominate the compute-cost expressions. Neglecting lower-order terms and equating the two costs therefore yields

$$\frac{1}{2} \Omega_{\text{CoT}}^2 \approx \frac{1}{2b} \Omega_{\text{ToT(1)}}^2 \Rightarrow \Omega_{\text{ToT(1)}} \approx \sqrt{b} \Omega_{\text{CoT}}. \quad (3.6)$$

Table 1. Comparison of various inference strategies: chain-of-thought (CoT), tree-of-thought (1) (ToT(1)), best-of- N (Bo N) and majority voting (MV). See electronic supplementary material, SI, for derivations.

	task success probability (ψ) ^a	$C_{\text{inf}}^{\text{param}}$	$C_{\text{inf}}^{\text{attn}}$	$C_{\text{inf}}^{\text{eval}} \mathbf{b}$	$\mathbb{E}[\Omega' \Omega]^c$
CoT	$l_p(m, \frac{\Omega}{\omega} - m + 1)$	$2N\Omega$	$2n_{\text{layer}}d_{\text{attn}}(\Omega_{\text{prompt}}^2 + \Omega_{\text{prompt}} + \frac{1}{2}\Omega^2)$	no	$\min\{\frac{\omega m}{l_p}, \Omega\}$
ToT(1)	$l_{1-(1-l_p)^b}(m, \frac{\Omega}{b\omega} - m + 1)$	$2N\Omega$	$2n_{\text{layer}}d_{\text{attn}}(\Omega_{\text{prompt}}^2 + \Omega_{\text{prompt}} + \frac{1}{2b}\Omega^2)$	yes	$\min\{\frac{\omega bm}{1-(1-l_p)^b}, \Omega\}$
Bo N ^d	$1 - \left(1 - l_p(m, \frac{\Omega}{k\omega} - m + 1)\right)^k$	$2N\Omega$	$2n_{\text{layer}}d_{\text{attn}}(k\Omega_{\text{prompt}}^2 + \Omega_{\text{prompt}} + \frac{1}{2k}\Omega^2)$	yes	$\min\{\frac{\omega km}{l_p}, \Omega\}$
MV ^d	$\mathbb{E}\left[\frac{\mathbb{1}\{Y_0 = \max_j Y_j\}}{1 + \sum_{j=1}^N \mathbb{1}\{Y_0 = Y_j\}}\right]$	$2N\Omega$	$2n_{\text{layer}}d_{\text{attn}}(k\Omega_{\text{prompt}}^2 + \Omega_{\text{prompt}} + \frac{1}{2k}\Omega^2)$	yes	$\min\{\frac{\omega km}{l_p}, \Omega\}$

^a $l_x(a, b)$ is the regularized incomplete beta function. MV is given over a distribution which includes incorrect answers such that $Y \sim \text{Multinomial}(k; \Pr[Y_0], \Pr[Y_1], \dots, \Pr[Y_{|Y|}])$ where the first entry is the probability of the correct answer $\Pr[Y_0] = l_p(m, \frac{\Omega}{k\omega} - m + 1)$ and the remaining $\Pr[Y_j]$ describe the mass assigned to incorrect outputs in the answer alphabet.

^bEvaluation costs may be significant for parallel methods such as ToT and Bo N but are negligible for majority voting.

^cPiecewise linear approximation of the expected tokens used $\mathbb{E}[\Omega' | \Omega]$ given a token budget Ω .

^dWe report the pass@ k for Bo N and MV with a CoT base policy such that the k trials each of length Ω/k independently process the prompt and proceed through their traces.

This scaling relationship demonstrates that ToT(1) can generate more tokens under a fixed compute budget, increasing with the branching factor. As shown in the forthcoming theorem 3.1, the preferred inference strategy varies with the task–capability trade-off, jointly determined by the number of required skills and the difficulty of executing each step. Regions of advantage shift with branching factor, per-step success probability and number of required skills.

Set-up: Let $\psi^{(b)}$ denote the task–success probability under a branching inference policy ToT(1) with branching factor b , where $\hat{p}$ is the per-step skill success probability, and m is the number of required sequential skills. The special case $b = 1$ corresponds to a purely sequential CoT strategy.

Theorem 3.1 (Task–capability trade-off between sequential and branching inference). The preferred inference strategy depends on the product $\hat{p}$.

- In the high-capability regime ($\hat{p} \rightarrow 1^-$), CoT inference ($b = 1$) minimizes the number of output tokens required for successful task completion and is therefore preferred.
- In the low-capability regime ($\hat{p} \rightarrow 0^+$), ToT(1) with branching factor b achieves a multiplicative gain in task success probability over CoT, with an advantage scaling as $b^{m/2}$ under a sufficiently large token budget.

This trade-off reflects the shifting advantage between sequential and branching strategies depending on model capability and task difficulty in conjunction with compute costs. The proof is provided in the electronic supplementary material, SI.

Conjecture 3.1 (Task–capability trade-off for branching inference: intermediate regime and large- m dominance). Motivated by numerical analysis (see electronic supplementary material, SI), we conjecture that there exists an optimal branching strategy for ToT(1), including CoT at $b = 1$, which jointly depends on the per-step success probability $\hat{p}$, task length m , branching factor b and token budget Ω_{budget} .

Fix any branching factor $b \geq 1$ and constant $c > \sqrt{b}$ (such that the shared budget scales as $\Omega_{\text{budget}} = cm\omega$). Then,

- For every $m > 1$, the function $\Delta(\varepsilon, m) = \psi^{(b)} - \psi^{(1)}$ is continuous in $\varepsilon \in [0, 1]$ and changes sign, guaranteeing a crossover $\varepsilon_*(m) \in (0, 1)$.
- Define $\varepsilon^\dagger = \sup\{\varepsilon : f_T(\varepsilon) < f_C(\varepsilon)\} \in (0, 1)$. For every fixed $\varepsilon < \varepsilon^\dagger$, there exists $m_0(\varepsilon)$ such that $\psi^{(b)} > \psi^{(1)}$ for all $m \geq m_0(\varepsilon)$; in fact, $\psi^{(b)} - \psi^{(1)} \rightarrow 1$ as $m \rightarrow \infty$.

CoT is preferred for $\varepsilon > \varepsilon_*(m)$; ToT(1)(b) is preferred for $\varepsilon < \varepsilon_*(m)$ and wins unconditionally once m is large relative to the fixed $\varepsilon < \varepsilon^\dagger$.

In practice, even under a fixed token budget and differing compute, sequential approaches are not always preferable to parallel ones [54,55]. This phenomenon can be understood within our framework by introducing fatigue or forgetting. Consider a pedagogical example with complete fatigue: for any chain that fails to yield a positive outcome at the first step (with success probability $r_1 = \hat{t}_1 p_1$), the success probability immediately collapses to zero on all subsequent steps, $r_{t>1} = 0$, owing to catastrophic error. In this case, without branching, the probability of success is exactly r_1 , since no further steps can compensate for failure. For strategies with b parallel traces, however, the probability becomes $1 - (1 - r_1)^b > r_1$ for all $b > 1$ and $0 < r_1 < 1$, so the preferred strategy is to fully parallelize the step budget. The relative effectiveness of sequential versus parallel inference varies across real systems, shaped by temporal dynamics in reasoning capabilities and task-specific dependencies, highlighting the value of adaptive strategies for improving both efficiency and robustness—particularly when task difficulty (as determined by the $(m, \hat{p})$ pair) interacts with fatigue or degradation.

Indeed, simpler reasoning strategies often outperform more complex strategies—particularly on easier tasks or those susceptible to bias [40,54,55,58]. Similar effects are observed in humans: individuals with higher numeracy may over-reason and reinforce prior beliefs [59]. In LLMs, bias

can arise from verifiers [28,46] or from internal representations [60], which we conjecture may lead to information cascades [61] that converge on incorrect outcomes. CoT reasoning has been shown to be more bias-prone and to underperform shorter chains in various settings [39,40]. Related observations have been made using rejection sampling methods that favour shorter completions [29], though this may reflect the tendency for correct reasoning to terminate early, while incorrect paths are longer and more exploratory, thereby consuming more of the token budget.

Wu *et al.* complemented their experiments with a stylized formulation to demonstrate that longer reasoning chains might degrade performance [40]. In their formulation, task success is decomposed into two multiplicative terms: the probability of posing correct sub-questions and the probability of answering them correctly. Specifically, the overall success probability over T steps and task complexity ℓ is approximated by $(1 - \sigma' \cdot \ell)^T (1 - E' \cdot \ell / T)^T \alpha$, where $\sigma' \cdot \ell$ reflects sub-question error, $E' \cdot \ell / T$ captures sub-answer error and α is a shortcut term representing end-to-end task solvability without decomposition. While highly simplified, this formulation captures an essential trade-off: deeper decompositions mitigate per-step difficulty but compound error across steps. A similar pattern emerges in a limiting case of the idealized DS3 framework when the inference budget is set equal to the number of steps, $T_{\max} = m$, and skill success probability becomes $\hat{p} = (1 - \sigma' \cdot \ell)(1 - E' \cdot \ell / m) \alpha^{1/m}$, exhibiting the same tension between solution depth and cumulative risk. This motivates a hierarchical view of task completion, where a model may either solve a problem using a small number of complex skills or a longer sequence of simpler ones. These results underscore the importance of evaluating inference strategies not in isolation, but in the context of how models are trained. The DS3 framework offers an analytical foundation for comparing reasoning strategies—not only in terms of their runtime behaviour but also in how structural priors are shaped during training—a perspective to which we now turn.

4. Unifying training and inference scaling in large language models

Gaining a deeper understanding of inference strategies and optimal resource allocation begins with examining the structure of the pretrained latent skill graph on which DS3 operates. To this end, we unify DS3 with a first-principles hierarchical tripartite graph framework and, in parallel, incorporate empirically grounded methods for characterizing LLM training. Analysing a single task in isolation is generally insufficient to capture broader performance trends. We therefore define a task distribution over the number of required skills and their difficulties, yielding direct expressions for the expected accuracy and corresponding expected cost.

(a) Accuracy scaling with training and inference compute

Consider the hierarchical skill-text tripartite graph framework for LLM training introduced in [18], which itself builds on [21,22,62]. We extend it to support flexible problem-solving strategies, including both upward and downward traversal through the skill hierarchy. In addition, we slightly modify the framework's functional form to better reflect empirical trends—specifically, the observation that overtraining a fixed-size model leads to improved but diminishing performance gains. This refinement is especially relevant in light of inference-time preferences that increasingly favour overtraining. Crucially, these adjustments do not alter the framework's core conclusions, including the Chinchilla-style scaling laws for compute-optimal training, the emergence of capabilities or the plateauing behaviour observed in LLMs as shown in [18].

Recent observations suggest that reasoning performance may scale predictably with resource allocation. Notably, linear accuracy improvements have been observed in reasoning models with logarithmic increases in training compute, inference compute and inference tokens [63,64]. We simulate task performance using a rounded and interpretable set of parameter values, selected to reflect plausible values without overfitting. The resulting trends are compared with publicly released results for Anthropic's Claude 3.7 Sonnet model [63] and OpenAI's o1 model [64] on the

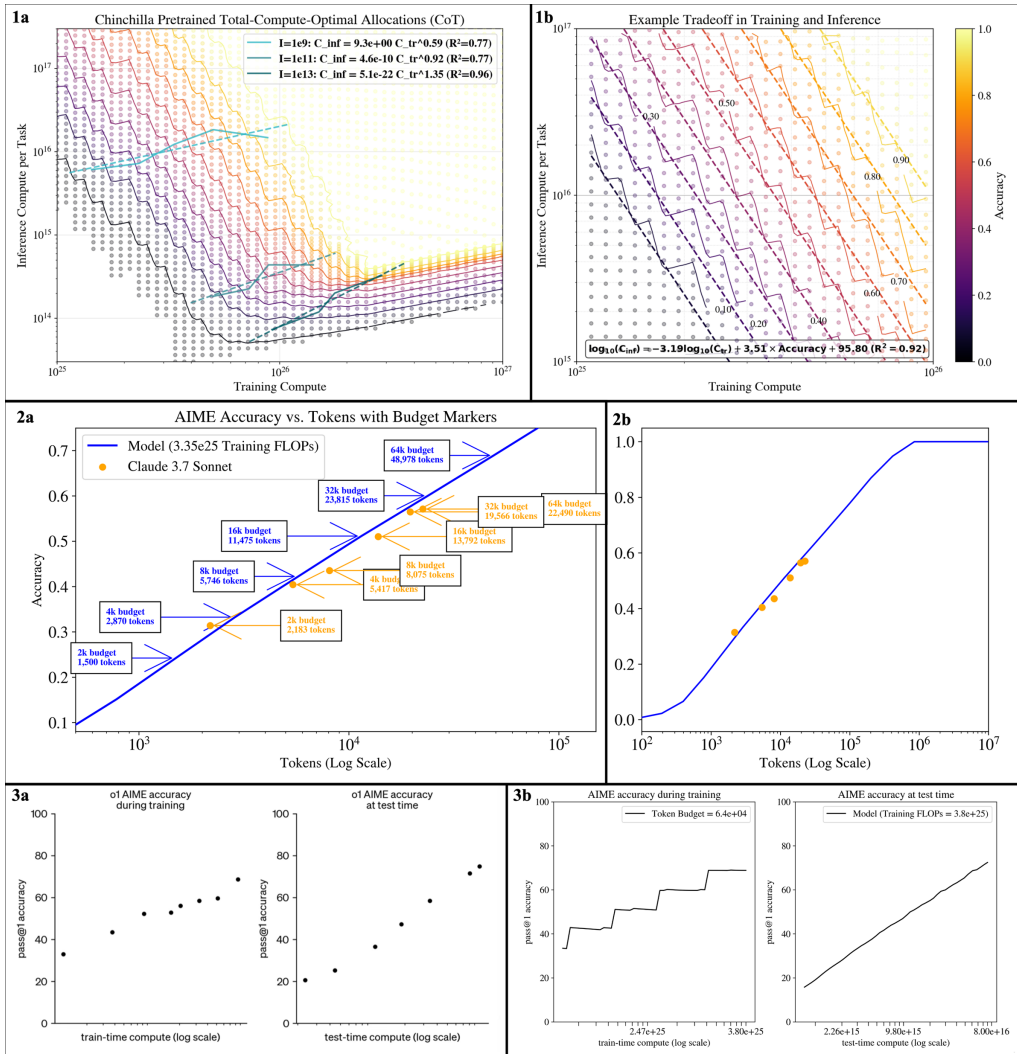


Figure 2. 1a. Contour plot of model accuracy under Chinchilla-style pretraining, showing total-compute-optimal allocations for a CoT policy (parameters aligned with Claude 3.7 Sonnet [63]). Inference compute excludes attention cost. 1b. Log-linear trade-offs between inference and pretraining compute across performance iso-contours. 2a–b. Accuracy scaling of Claude 3.7 on AIME versus log tokens used per task, adapted from [63], using EpochAI-estimated training FLOPs [65]. 3a. Scaling behaviour of OpenAI's o1 on AIME in training and inference compute [66]. 3b. Reproduction using larger training data estimates [65] with a 64k-token inference budget.

American Invitational Mathematics Examination (AIME). Optimal scaling behaviour shifts with the number of downstream tasks. As shown in figure 2.1a, models should allocate proportionally more compute to pretraining when amortized over a larger task load. This shift also modifies the total-compute-optimal scaling laws. The plotted regimes reflect realistic, rounded parameters. Since these are general-purpose LLMs, the precise allocation ultimately depends on the task mix.

A logarithmic trade-off between training and inference compute has been observed in games such as Hex, where performance iso-contours indicate increased inference can compensate for reduced training and vice versa [67]. Similar trade-offs have been considered more broadly, including in LLMs, where meta-analyses show that increasing inference compute by orders of magnitude can offset training cost—though the optimal ratio varies with model scale and task difficulty [54,55,68]. Figure 2.1b illustrates the core trade-off: compute can be allocated to training or

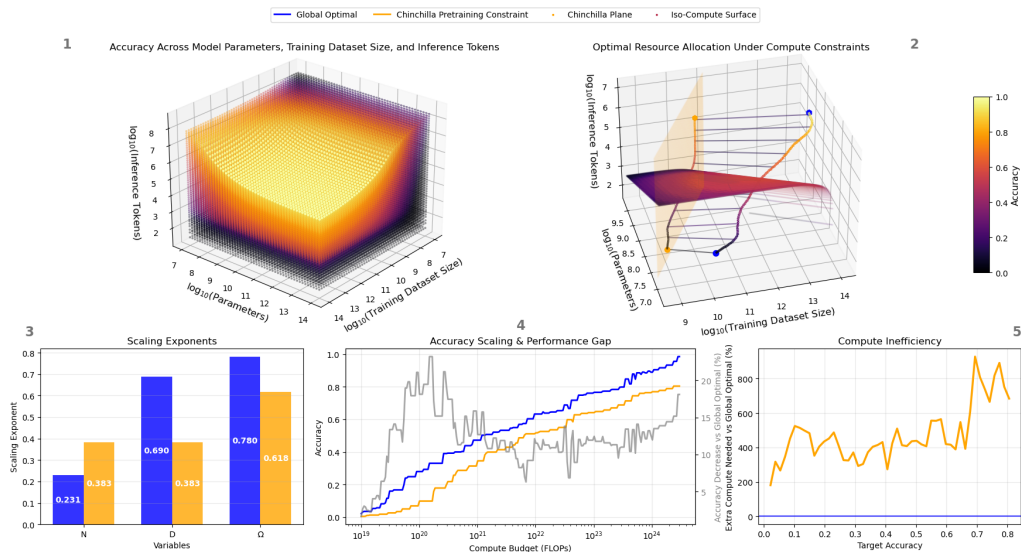


Figure 3. Hypothetical comparison of global optimal versus Chinchilla pretrained. (1) Log-space accuracy surface used for optimization. (2) Optimal trajectories on a constant-compute surface: unconstrained global optimum, Chinchilla pretrained and plane ($\frac{D}{N} = 20$). (3) Fitted power-law exponents across compute. (4) Chinchilla trails the global path in accuracy. (5) Matching accuracy costs significantly more compute.

inference, and performance iso-contours are approximately log-linear in this space. Figure 2.2a–3b further show that accuracy scales linearly with the logarithm of training compute, inference compute and tokens used, reinforcing the empirical regularities predicted by our framework.

(b) Efficiency benefits of inference scaling

Maximizing performance relative to resource expenditure is a central imperative for sustainable AI and efficient system design. This motivates a closer examination of where inefficiencies emerge in standard model scaling practices. To demonstrate how DS3 can be integrated with empirical methodologies for analysing LLM training, we begin with a widely used empirical model of pre-training loss [2], which serves as the foundation for estimating a model's learnt competence. A two-step approach is used to connect pretraining loss with downstream task performance. First, we map pretraining loss to skill-level proficiency via sigmoid transformation, following recent work on scaling-to-task transfer [24].

As illustrated in the synthetic scenario of figure 3, constraining optimization to the two-dimensional plane defined by Chinchilla-style training compute optimality can lead to substantial inefficiency. Our analysis indicates that similar limitations arise in strategies that optimize parameter count and dataset size under a fixed inference token budget, as in [51], since such methods restrict scaling to a different two-dimensional slice of the broader design space (see electronic supplementary material, SI, figure S7). In contrast, allowing joint optimization over model size N , dataset size D and inference token budget Ω provides access to significantly more efficient scaling regimes.

For inference-compute-optimal scaling, the dataset size does not affect the cost. Therefore, we consider the regime in which data are effectively unbounded to analyse the relationship between model parameters and inference tokens. We analyse this setting across several orders of magnitude, focusing on individual tasks as defined by their difficulty. The findings suggest that tasks requiring more complex skills demand higher base model capability and therefore larger model size, but benefit primarily from scaling inference tokens. In contrast, tasks involving simple or

moderately difficult skills exhibit scaling in both parameters and tokens, with base requirements that vary according to task complexity.

(c) Can reasoning elicit emergence?

Increasing the depth of reasoning appears to lower the model size at which new capabilities emerge. Empirically, this has been observed in CoT prompting, where deeper reasoning shifts flat or even inverse scaling in model size into positive scaling [32,33].

Theorem 4.1 (Reasoning depth amplifies model scaling benefits). *Let $S(N, t) = 1 - (1 - r(N))^t$ denote the probability of successfully completing a skill in t reasoning steps, where $r(N) \in (0, 1)$ is a strictly increasing function (i.e. $dr/dN > 0$) representing the per-step success probability of a model with N non-embedding parameters. Define the sensitivity of accuracy to model size as the N -slope,*

$$\Delta(N, t) \frac{\partial S}{\partial N} = t(1 - r(N))^{t-1} \frac{dr}{dN}.$$

Consider how $\Delta(N, t)$ changes with reasoning depth t at a model size N , focusing on the regime when $r(N)$ is small and the slope $\Delta(N, 1)$ is shallow, corresponding to model sizes where accuracy has plateaued at low reasoning depth. In this setting, increasing t steepens the slope, improving scaling. See the proof in the electronic supplementary material, SI.

We identify three conceptual mechanisms through which reasoning depth enhances performance when accuracy has plateaued with increasing model size.

(i) *Model size-to-skill capability mapping.* Theorem 4.1 demonstrates that, for a model size N where performance has plateaued, increasing reasoning depth t not only increases task success probability $S(N, t)$ but also steepens its derivative with respect to N —that is, accuracy improves more sharply with model size under deeper reasoning.

(ii) *Skill-count bottleneck.* When a task requires m skill applications that are currently available to the model, no number of reasoning steps $t < m$ suffices for task completion. Therefore, the model must scale in size to acquire new skills. Yet, doing so may require a substantial increase in training. Increasing t beyond m removes this bottleneck, enabling continued scaling by improving the proficiency of existing skills required for the task.

(iii) *Decomposition.* Reasoning steps may allow the model to decompose complex tasks into sequences of simpler sub-skills, effectively reducing per-step difficulty. This allows structured inference to compensate for limited base competence, while facilitating continued performance gains with depth, analogous to mechanism (ii) but driven by internal task restructuring.

These mechanisms independently reproduce the qualitative pattern in which increased reasoning depth leads to earlier gains in capability. When applied across a distribution of task difficulties, the framework predicts two characteristic behaviours: for task sets with broad and continuous difficulty distributions, performance improves smoothly as the number of reasoning steps increases; for task sets with discretely spaced or clustered difficulty levels, performance may exhibit sharp, step-like emergence curves. Under a task-centric interpretation of emergence [18], this suggests that reasoning can actively elicit emergent behaviour, shifting capability thresholds to smaller scales. In both settings, additional reasoning can enable smaller models to solve tasks that would otherwise remain unsolvable. To push performance further, one may also leverage aggregation techniques operating over multiple reasoning samples. This motivates a closer study of BoN and MV behaviour across heterogeneous task populations.

(d) Best-of- N coverage and majority vote saturation

For mixed populations of tasks with varying base success rates, BoN continues to show meaningful gains with additional samples, while MV tends to saturate more quickly over practical sampling budgets. Separate works have characterized repeated sampling performance in

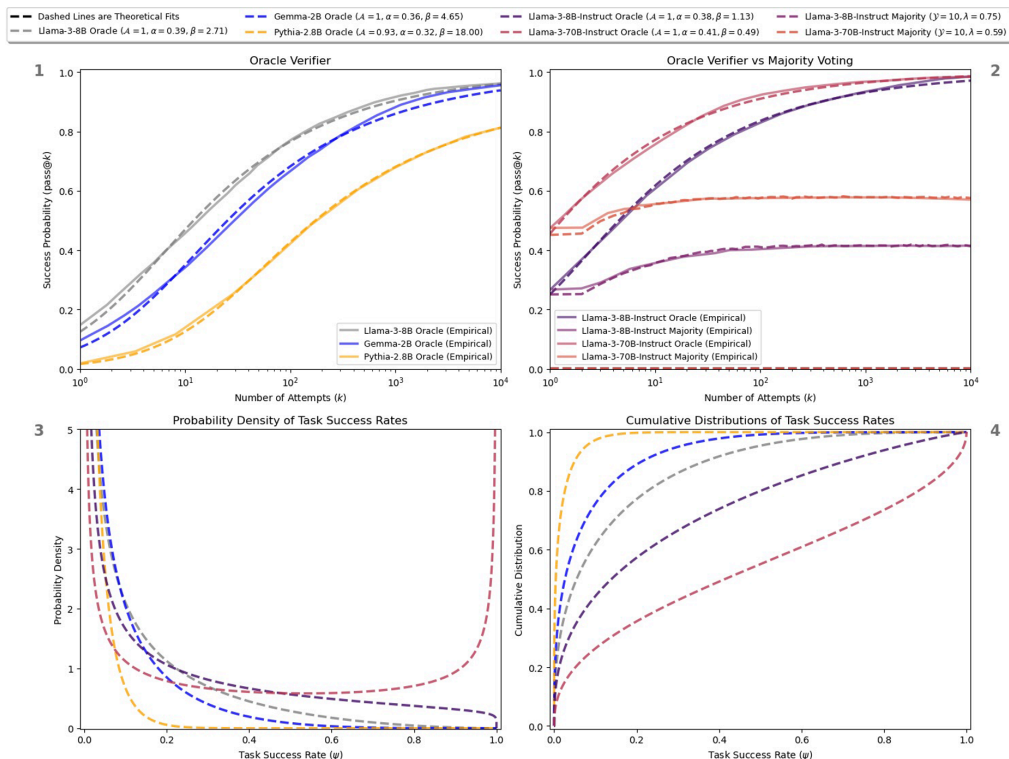


Figure 4. Comparison of oracle verifier and majority vote (MV) accuracy using empirical pass@k and beta-distributed task success rates. Solid lines show empirical data from [71] on the MATH benchmark [72]. (1) Pass@k for BoN with oracle verifier on three base models; dashed lines are beta-fit predictions [70]. (2) Same oracle curves (solid) versus MV accuracy (dashed) for two instruction-tuned Llama-3 models, showing early MV saturation. (3) Probability density functions of task success rates from beta fits in (1) and (2), showing mass near zero and, for Llama-3-70B (peach), a secondary high-skill peak. (4) Corresponding cumulative distribution functions illustrating increasing spread with model capability.

terms of population-level heterogeneity in pass or error rates [69], including models that assume beta-distributed task difficulty [70]. Saturation in MV has been explained through per-instance convergence dynamics [55].

A recent theoretical formulation for pass@k proposed by Levi *et al.* [70] treats the error rate as a beta-distributed random variable, which asymptotically yields a power-law decay in failure probability (formally derived in electronic supplementary material, SI). Motivated by this observation, we represent the induced success probability distribution as a mixture of a beta distribution and a point mass at zero, representing tasks the model cannot solve. This recovers the work of Levi *et al.* and closely matches empirical pass@k curves reported by Brown *et al.* [71] on the MATH benchmark [72] for the Pythia-2.8B [73], Llama-3-8B [24] and Gemma-2B [74] models.

We perform our own fit under this distribution, reporting slightly different best-fit parameters, and visualize the resulting curves in figure 4, including the predicted cumulative distribution function and probability density function of task-level success probabilities. In addition, we extend the analysis to two other models: Llama-3-8B-Instruct and Llama-3-70B-Instruct [24]. Across all models, we observe a pronounced mass of tasks with low success probability. For Llama-3-70B-Instruct, we find a secondary peak near perfect success rates, indicating a subset of tasks that are probably trivial for the larger model. The same framework used for BoN can also capture MV scaling behaviour as a function of the number of steps. To demonstrate this, we represent the probability of incorrect answers with a truncated geometric distribution. This form captures a decaying preference over incorrect outputs, a plausible behaviour for language models, which tend

to concentrate probability mass on a few high-likelihood completions even when incorrect [27]. We find that the 70B-Instruct model has a more concentrated distribution than the 8B-Instruct. One possible interpretation is that larger models, which tend to achieve lower pretraining loss, not only increase confidence in correct predictions but also exhibit sharper preferences among incorrect ones—an effect reminiscent of overtrained models, as discussed in [27]. The final fit to MV accuracy is shown in figure 4.2.

As a proof of concept, we further demonstrate that a distribution over task-level parameters can be pushed forward through the two-step sigmoid ansatz to estimate the model-specific search directionality parameter $\hat{\iota}$. Despite strong simplifying assumptions—including a two-mode Gaussian task prior and the exclusion of alternative solution strategies—the estimated values of $\hat{\iota}$ are qualitatively consistent with expectations, assigning higher directionality to instruction-tuned models. Since this construction is inherently sensitive to the assumed task distribution and has not yet undergone a thorough robustness analysis, we leave a more detailed investigation to future work.

5. Conclusion

As LLMs continue to advance, understanding and strategically allocating compute across training and inference has become critical for maximizing performance and efficiency. In this work, we presented a unified framework for compute scaling that integrates both training and inference within a shared mathematical structure. By framing reasoning as a directed stochastic search over a latent skill graph, we demonstrated how key empirical phenomena naturally arise from the interaction between inference dynamics, training structure and task difficulty. Our analysis considers inference as a foundational axis of model capability—one that increasingly calls for co-design across training and inference to better align algorithms, system architectures and hardware platforms throughout the full lifecycle of model development and deployment.

As inference becomes an increasingly dominant use of energy, aligning system performance with dynamic energy availability and usage patterns demands new principles of system-level design. The computational and energy dynamics of LLMs reveal key differences between training and inference, especially in deployment contexts where energy, latency and hardware constraints vary significantly. Inference efficiency is shaped by architectural choices (e.g. mixture-of-experts [75], multi-query attention [76]) and reasoning strategies. Future principles may include probabilistic precomputing to store energy in information [77], intelligent routing by task type [78] and difficulty and tranching intelligence governed by renewable energy availability [79]. Security risks scale with unfiltered inference capabilities, but mitigation methods can reduce risk with further inference security measures [80,81]. The interplay between capability, interpretability and responsiveness remains complex and unresolved with emerging regulatory and governance challenges as reasoning models become more distributed and decentralized [82]. A clearer understanding of compute dynamics can help guide the design of models, infrastructure and policies that are more efficient and better aligned with societal needs and constraints. Moreover, these dynamics give rise to qualitatively distinct optimization regimes, each with its own characteristic scaling behaviour.

As in biological systems, increasing scale alters a system's functional demands [49]. When organisms grow larger, they often shift ecological roles and constraints—what was optimal at one scale becomes suboptimal at another. Scaling modifies not only resource trade-offs but also the broader integration of AI into social, economic and technical ecosystems, creating a feedback loop in which new capabilities reshape the underlying conditions that govern further scaling.

Data accessibility. Accessible from <https://github.com/austine4/Inference-Compute-Scaling-Laws>.

Supplementary material is available online [83].

Declaration of AI use. AI was used as a copilot in coding and to edit language.

Authors' contributions. A.R.E.-M.: conceptualization, data curation, formal analysis, investigation, methodology, software, visualization, writing—original draft, writing—review and editing; A.K.N.: conceptualization, formal analysis, investigation, methodology, visualization, writing—review and editing; L.R.V.: conceptualization, funding acquisition, investigation, project administration, supervision, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

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